Chapter 9

Campbell Biology, AP Edition, 12e

A hoary marmot eats a flower.

hidecaption

Chapter 9

Cellular Respiration and Fermentation

Figure 9.1

This hoary marmot (*Marmota caligata*) obtains energy for its cells by feeding on plants. In the process of cellular respiration, mitochondria in the cells of animals, plants, and other organisms break down organic molecules, generating ATP and waste products: carbon dioxide, water, and heat. Note that energy flows one way, but chemicals are recycled.

How is the chemical energy stored in food used to generate A T P, the molecule that drives most cellular work?

Light energy is used in photosynthesis. It generates organic molecules plus O 2, which are used in plants the marmot eats. This generates C O 2 plus H 2 O, which are used in photosynthesis, creating a cycle. When the marmot eats, cellular respiration in mitochondria occurs in plant cells and animal cells. This breaks down organic molecules, generating A T P, which powers most cellular work. A T P releases heat.

Key Concepts

- **9.1Catabolic pathways yield energy by oxidizing organic fuels****
- **9.2Glycolysis harvests chemical energy by oxidizing glucose to pyruvate****
- **9.3After pyruvate is oxidized, the citric acid cycle completes the energy-yielding oxidation of organic molecules****

- **9.4During oxidative phosphorylation, chemiosmosis couples electron transport to ATP synthesis****
- **9.5Fermentation and anaerobic respiration enable cells to produce ATP without the use of oxygen****
- **9.6Glycolysis and the citric acid cycle connect to many other metabolic pathways****

Advanced placement edition

Heterotrophic organisms must capture energy from their environment to maintain life **(BIG IDEA 2).** Common to all are core processes that move energy and matter through living systems and thereby support life **(BIG IDEAS 1 & 4).**

Study Tip

Make a visual study guide: Draw a cell with a large mitochondrion, labeling the parts of the mitochondrion. As you go through the chapter, add key reactions for each stage of cellular respiration, linking the stages together. Label the carbon molecule(s) with the most energy and the carbon molecule(s) with the least energy. Your cell can be a simple sketch, as shown here.

An illustration of a mitochondrion, showing the double-membraned structure with inward loops.

Get Ready for This Chapter

In this chapter, you will learn about the processes involved in using chemical energy stored in food to make ATP, a molecule that can be used as a source of energy for cellular processes. Before beginning this chapter, you should be able to do the following:

- Describe the free energy changes in exergonic and endergonic reactions (see Concept 8.2, section entitled “Free Energy and Metabolism”).
- Explain the role of ATP in metabolism (see Concept 8.3).
- Describe the structure of a fat and a carbohydrate (see Concept 5.2 and Concept 5.3).
- Explain the potential energy of electrons (see Concept 2.2).
- Explain the relationship between electronegativity and the polarity of a covalent bond (see Concept 2.3).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following is true for all exergonic reactions?

```
A net input of energy from the surroundings is required for the reactions to proceed.
```

That's incorrect. Try again.

Incorrect. Review the following material: Chemical reactions are classified based on their free-energy changes. An exergonic reaction proceeds with a net release of free energy, whereas an endergonic reaction is one that absorbs free energy from its surroundings. If a chemical process is exergonic (downhill), releasing energy in one direction, then the reverse process must be endergonic (uphill), using energy.

```
The products have more total energy than the reactants.
```

That's incorrect. Try again.

Incorrect. Review the following material: Chemical reactions are classified based on their free-energy changes. An exergonic reaction proceeds with a net release of free energy, whereas an endergonic reaction is one that absorbs free energy from its surroundings. If a chemical process is exergonic (downhill), releasing energy in one direction, then the reverse process must be endergonic (uphill), using energy.

```
The reaction goes only in a forward direction: All reactants will be converted to products.
```

That's incorrect. Try again.

Incorrect. Review the following material: Chemical reactions are classified based on their free-energy changes. An exergonic reaction proceeds with a net release of free energy, whereas an endergonic reaction is one that absorbs free energy from its surroundings. If a chemical process is exergonic (downhill), releasing energy in one direction, then the reverse process must be endergonic (uphill), using energy.

```
The reaction proceeds with a net release of free energy.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Chemical reactions are classified based on their free-energy changes. An exergonic reaction proceeds with a net release of free

energy, whereas an endergonic reaction is one that absorbs free energy from its surroundings. If a chemical process is exergonic (downhill), releasing energy in one direction, then the reverse process must be endergonic (uphill), using energy.

Energy from catabolism provides the energy to form ATP from ADP plus Pi. The breakdown of ATP to ADP plus Pi provides energy for cellular work.

2. Which of the following is the most correct interpretation of the figure shown?

Pi acts as a shuttle molecule to move energy from ATP to ADP.

That's incorrect. Try again.

Incorrect. Review the following material: ATP is a renewable resource that can be regenerated by the addition of inorganic phosphate Pi to ADP. Energy released by catabolism in the cell is used to phosphorylate ADP. regenerating ATP. Chemical energy stored in ATP drives most cellular work.

```
        Energy from catabolism can be used directly for
performing cellular work.
```

That's incorrect. Try again.

Incorrect. Review the following material: ATP is a renewable resource that can be regenerated by the addition of inorganic phosphate Pi to ADP. Energy released by catabolism in the cell is used to phosphorylate ADP, regenerating ATP. Chemical energy stored in ATP drives most cellular work.

ATP is a molecule that acts as an intermediary to store chemical energy for cellular work.

That's correct.

Correct! If you had trouble with this question, review the following material: ATP is a renewable resource that can be regenerated by the addition of inorganic phosphate (Pi) to ADP. Energy released by catabolism in the cell is used to phosphorylate ADP, regenerating ATP. Chemical energy stored in ATP drives most cellular work.

ADP and Pi are a set of molecules that store energy for catabolism.

That's incorrect. Try again.

Incorrect. Review the following material: ATP is a renewable resource that can be regenerated by the addition of inorganic phosphate Pi to ADP. Energy released by catabolism in the cell is used to phosphorylate ADP, regenerating ATP. Chemical energy stored in ATP drives most cellular work.

3. Two molecules with the chemical formulas $C_6H_{12}O_6$ and $C_6H_{12}O_2$ are probably

```
a monosaccharide and a fatty acid
```

That's correct.

Correct! If you had trouble with this question, review the following material: Monosaccharides generally have molecular formulas that are some multiple of the unit CH_2O. Glucose ($C_6H_{12}O_6$) is the most common monosaccharide. A fatty acid has a long carbon skeleton with a carbon at one end that is part of a carboxyl group (-COOH); the remainder of the skeleton consists of a hydrocarbon chain. The relatively nonpolar C-H bonds in the hydrocarbon chains of fatty acids are the reason fats are hydrophobic.

```
a monosaccharide and a nucleotide
```

That's incorrect. Try again.

Incorrect. Review the following material: Monosaccharides generally have molecular formulas that are some multiple of the unit CH_2O. Glucose ($C_6H_{12}O_6$) is the most common monosaccharide. A fatty acid has a long carbon skeleton with a carbon at one end that is part of a carboxyl group -COOH; the remainder of the skeleton consists of a hydrocarbon chain. The relatively nonpolar C-H bonds in the hydrocarbon chains of fatty acids are the reason fats are hydrophobic.

```
a nucleic acid and a fat
```

That's incorrect. Try again.

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```
a fat and a nucleotide
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a long carbon skeleton with a carbon at one end that is part of a carboxyl group (-COOH); the remainder of the skeleton consists of a hydrocarbon chain. The relatively nonpolar C-H bonds in the hydrocarbon chains of fatty acids are the reason fats are hydrophobic.

4. Which of the following is true of the energy levels of electrons in shells?

```
     Valence electrons have a higher energy level than
those in other filled shells.
```

That's correct.

Correct! If you had trouble with this question, review the following material: The negatively charged electrons of an atom are attracted to the positively charged nucleus. It takes work to move a given electron farther away from the nucleus, so the more distant an electron is from the nucleus, the greater its potential energy. The valence electrons, located in the outermost electron shell (the valence shell), have the highest energy level.

```
        Electrons must lose energy to move from the first to the
second shell.
```

That's incorrect. Try again.

Incorrect. Review the following material: The negatively charged electrons of an atom are attracted to the positively charged nucleus. It takes work to move a given electron farther away from the nucleus, so the more distant an electron is from the nucleus, the greater its potential energy. The valence electrons, located in the outermost electron shell (the valence shell), have the highest energy level.

```
        All the electrons in an atom have similar energy levels.
```

That's incorrect. Try again.

Incorrect. Review the following material: The negatively charged electrons of an atom are attracted to the positively charged nucleus. It takes work to move a given electron farther away from the nucleus, so the more distant an electron is from the nucleus, the greater its potential energy. The valence electrons, located in the outermost electron shell (the valence shell), have the highest energy level.

```
        Electrons do not have potential energy, just kinetic
energy.
```

That's incorrect. Try again.

Incorrect. Review the following material: The negatively charged electrons of an atom are attracted to the positively charged nucleus. It takes work to move a given electron farther away from the nucleus, so the more distant an electron is from the nucleus, the greater its potential energy. The valence electrons, located in the outermost electron shell (the valence shell), have the highest energy level.

5. A covalent bond is likely to be polar when

```
     the two atoms sharing electrons are different
elements.
```

That's incorrect. Try again.

Incorrect. Review the following material: When an atom is bonded to a more electronegative atom, the electrons of the bond are not shared equally and a polar covalent bond is formed. Such bonds vary in their polarity, depending on the relative electronegativity of the two atoms. Atoms that have similar (or the same) electronegativities will form a nonpolar covalent bond. For example, a C-H bond is nonpolar because C and H have similar electronegativities.

```
        the two atoms sharing electrons have the same
electronegativity.
```

That's incorrect. Try again.

Incorrect. Review the following material: When an atom is bonded to a more electronegative atom, the electrons of the bond are not shared equally and a polar covalent bond is formed. Such bonds vary in their polarity, depending on the relative electronegativity of the two atoms. Atoms that have similar (or the same) electronegativities will form a nonpolar covalent bond. For example, a C-H bond is nonpolar because C and H have similar electronegativities.

```
        the two atoms sharing electrons are the same elements.
```

That's incorrect. Try again.

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```
        one of the atoms sharing electrons is more
electronegative than the other.
```

That's correct.

Correct! If you had trouble with this question, review the following material: When an atom is bonded to a more electronegative atom, the electrons of the bond are not shared equally and a polar covalent bond is formed. Such bonds vary in their polarity, depending on the relative electronegativity of the two atoms. Atoms that have similar (or the same) electronegativities will form a nonpolar covalent bond. For example, a C-H bond is nonpolar because C and H have similar electronegativities.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 9.1Explain how the redox reactions of catabolic pathways yield energy by oxidizing organic fuels.
- 9.2Trace the pathway of glucose oxidation during glycolysis.
- 9.3Describe the oxidation of pyruvate and the process by which further oxidation occurs in the citric acid cycle.
- 9.4Identify the steps of oxidative phosphorylation and account for the total ATP produced per glucose molecule during cellular respiration.
- 9.5Name two types of fermentation and explain how they result in ATP production.
- 9.6Use examples to illustrate the interactions of glycolysis and the citric acid cycle with other metabolic pathways.

Concept 9.1: Catabolic pathways yield energy by oxidizing organic fuels

Living cells require transfusions of energy from outside sources to perform their many tasks—for example, assembling polymers, pumping substances across membranes, moving, and reproducing. The outside source of energy is food, and the energy stored in the organic molecules of food ultimately comes from the sun. As shown in **Figure 9.1**, energy flows into an ecosystem as sunlight and exits as heat; in contrast, the chemical elements essential to life are recycled. Photosynthesis generates oxygen, as well as organic molecules used by the mitochondria of eukaryotes as fuel for cellular respiration. Respiration breaks this fuel down, using oxygen O_2 and generating ATP. The waste products of this type of respiration, carbon dioxide CO_2 and water H_2O, are the raw materials for photosynthesis.

Animation

Energy Flow and Chemical Recycling

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Let's consider how cells harvest the chemical energy stored in organic molecules and use it to generate ATP, the molecule that drives most cellular work. Metabolic pathways that release stored energy by breaking down complex molecules are called catabolic pathways (see Concept 8.1). Transfer of electrons from food molecules (like glucose) to other molecules plays a major role in these pathways. In this section, we consider these processes, which are central to cellular respiration.

Catabolic Pathways and Production of ATP

Organic compounds possess potential energy as a result of the arrangement of electrons in the bonds between their atoms. Compounds that can participate in exergonic reactions can act as fuels. Through the activity of enzymes (see Concept 8.4), a cell systematically degrades complex organic molecules that are rich in potential energy to simpler waste products that have less energy. Some of the energy taken out of chemical storage can be used to do work; the rest is dissipated as heat.

One catabolic process, **fermentation**, is a partial degradation of sugars or other organic fuel that occurs without the use of oxygen. However, the most efficient catabolic pathway is **aerobic respiration**, in which oxygen is consumed as a reactant along with the organic fuel (*aerobic* is from the Greek *aer*, air, and *bios*, life). The cells of most eukaryotic and many prokaryotic organisms can carry out aerobic respiration. Some prokaryotes use substances other than oxygen as reactants in a similar process that harvests chemical energy without oxygen; this process is called *anaerobic respiration* (the prefix *an-* means "without"). Technically, the term **cellular respiration** includes both aerobic and anaerobic processes. However, it originated as a synonym for aerobic respiration because of the relationship of that process to organismal respiration, in which an animal breathes in oxygen. Thus, *cellular respiration* is often used to refer to the aerobic process, a practice we follow in most of this chapter.

Although very different in mechanism, aerobic respiration is in principle similar to the combustion of gasoline in an automobile engine after oxygen is mixed with the fuel (hydrocarbons). Food provides the fuel for respiration, and the exhaust is carbon dioxide and water. The overall process can be summarized as follows:

$$\text{Organic compounds} + \text{Oxygen} \rightarrow \text{Carbon dioxide} + \text{Water} + \text{Energy}$$

Carbohydrates, fats, and proteins from food can all be processed and consumed as fuel. In animal diets, a major source of carbohydrates is starch, a storage polysaccharide that can be broken down into glucose $C_6H_{12}O_6$ subunits. Here, we will learn the steps of cellular respiration by tracking the degradation of the sugar glucose:

$$C_6H_{12}O_6 + 6\,O_2 \rightarrow 6\,CO_2 + 6\,H_2O + \text{Energy (ATP + heat)}$$

BioFlix® Animation

Introduction to Cellular Respiration

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This breakdown of glucose is exergonic, having a free-energy change of −686 kcal (-2,870 kJ) per mole of glucose decomposed ($\Delta G = -686$ kcal/mol). Recall that a negative ΔG ($\Delta G < 0$) indicates that the products of the chemical process store less energy than the reactants and that the reaction can happen spontaneously—in other words, without an input of energy (see Concept 8.2).

Catabolic pathways do not directly move flagella, pump solutes across membranes, polymerize monomers, or perform other cellular work. Catabolism is linked to work by a chemical drive shaft—ATP (see Concept 8.3). To keep working, the cell must regenerate its supply of ATP from ADP and Ⓟi (see Figure 8.12). To understand how cellular respiration accomplishes this, let's examine the fundamental chemical processes known as oxidation and reduction.

Redox Reactions: Oxidation and Reduction

How do the catabolic pathways that decompose glucose and other organic fuels yield energy? The answer is based on the transfer of electrons during the chemical reactions. The relocation of electrons releases energy stored in organic molecules, and this energy ultimately is used to synthesize ATP.

The Principle of Redox

In many chemical reactions, there is a transfer of one or more electrons e− from one reactant to another. These electron transfers are called oxidation-reduction reactions, or **redox reactions** for short. In a redox reaction, the loss of electrons from one substance is called **oxidation**, and the addition of electrons to another substance is known as **reduction**. (Note that *adding* electrons is called *reduction*; adding negatively charged electrons to an atom *reduces* the amount of positive charge of that atom.)

To take a simple, nonbiological example, consider the reaction between the elements sodium (Na) and chlorine (Cl) that forms table salt:

N a plus C l yields N a superscript plus, plus C l superscript minus. N a becomes oxidized (loses electron) as N a superscript plus. C l becomes reduced (gains electron) as C l superscript minus.

We could generalize a redox reaction this way:

X e minus plus Y yields X plus Y e minus. X e minus becomes oxidized as X. Y becomes reduced as Y e minus.

In the generalized reaction, substance Xe−, the electron donor, is called the **reducing agent**; it reduces Y, which accepts the donated electron. Substance Y, the electron acceptor, is the **oxidizing agent**; it oxidizes Xe− by removing its electron. Because an electron transfer requires both an electron donor and an acceptor, oxidation and reduction always go hand in hand.

Not all redox reactions involve the complete transfer of electrons from one substance to another; some change the degree of electron sharing in covalent bonds. Methane combustion, shown in **Figure 9.2**, is an example. The covalent electrons in methane are shared nearly equally between the bonded atoms because carbon and hydrogen have about the same affinity for valence electrons; they are about equally

electronegative (see Concept 2.3). But when methane reacts with O_2, forming CO_2, electrons end up shared less equally between the carbon atom and its new covalent partners, the oxygen atoms, which are very electronegative. In effect, the carbon atom has partially “lost” its shared electrons; thus, methane has been oxidized.

Figure 9.2

Methane combustion as an energy-yielding redox reaction.

C H 4 plus 2 O 2 goes to C O 2 plus Energy plus 2 H 2 O. C H 4 is oxidized into C O 2 and O 2 is reduced into H 2 O.

The reaction releases energy to the surroundings because the electrons lose potential energy when they end up being shared unequally, spending more time near electronegative atoms such as oxygen.

Electron sharing in covalent bonds is shown for each reactant and product. In methane, the reducing agent, carbon shares a pair of electrons equally in four covalent bonds with each of the four hydrogen atoms. In oxygen, the oxidizing agent, two pairs of electrons are equally shared in a covalent double bond between the two oxygen atoms. In carbon dioxide, carbon shares two pairs of electrons with each of two oxygen atoms, but the electrons are not shared equally and are pulled towards both oxygen atoms. In water, oxygen shares a pair of electrons with each of two hydrogen atoms, but the electrons are not shared equally and are pulled towards oxygen.

VISUAL SKILLS *Is the carbon atom oxidized or reduced during this reaction? Explain.*

*For suggested answer, see*Appendix A.**

Animation

Redox Reactions

image

Play

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Play

Now let’s examine the fate of the reactant O_2. The two atoms of O_2 share their electrons equally. But after the reaction with methane, when each O atom is bonded to two H atoms in H_2O, the electrons of those covalent bonds spend more time near the oxygen (see Figure 9.2). In effect, each O atom has partially

“gained” electrons, so the oxygen molecule O_2 has been reduced. Because the O atom is so electronegative, O_2 is one of the most powerful of all oxidizing agents.

Energy must be added to pull an electron away from an atom, just as energy is required to push a ball uphill. The more electronegative the atom (the stronger its pull on electrons), the more energy is required to take an electron away from it. An electron loses potential energy when it shifts from a less electronegative atom toward a more electronegative one, just as a ball loses potential energy when it rolls downhill. A redox reaction that moves electrons closer to an O atom, such as the burning (oxidation) of methane, therefore releases chemical energy that can be put to work.

Oxidation of Organic Fuel Molecules During Cellular Respiration

The oxidation of methane by O_2 is the main combustion reaction that occurs at the burner of a gas stove. The combustion of gasoline in an automobile engine is also a redox reaction; the energy released pushes the pistons. But the energy-yielding redox process of greatest interest to biologists is respiration: the oxidation of glucose and other molecules in food. Examine again the summary equation for cellular respiration, but this time think of it as a redox process:

C 6 H 12 O 6 plus 6 O 2 yields 6 C O 2 plus 6 H 2 O plus energy. C 6 H 12 O 6 becomes oxidized as 6 C O 2. 6 O 2 becomes reduced as 6 H 2 O.

As in the combustion of methane or gasoline, the fuel (glucose) is oxidized and O_2 is reduced. The electrons lose potential energy along the way, and energy is released.

In general, organic molecules that have an abundance of hydrogen are excellent fuels because their bonds are a source of “hilltop” electrons, whose energy may be released as these electrons “fall” down an energy gradient during their transfer to oxygen. The summary equation for respiration indicates that hydrogen is transferred from glucose to the O atoms in O_2. But the important point, not visible in the summary equation, is that the energy state of the electron changes as hydrogen (with its electron) is transferred to oxygen. In respiration, the oxidation of glucose transfers electrons to a lower energy state, liberating energy that becomes available for ATP synthesis. So, in general, we see fuels with multiple C—H bonds oxidized into products with multiple C—O bonds.

The main energy-yielding foods—carbohydrates and fats—are reservoirs of electrons associated with hydrogen, often in the form of C—H bonds. Only the barrier of activation energy holds back the flood of electrons to a lower energy state (see Figure 8.13). Without this barrier, a food substance like glucose would combine almost instantaneously with O_2. If we supply the activation energy by igniting glucose, it burns in air, releasing 686 kcal (2,870 kJ) of heat per mole of glucose (about 180 g). Body temperature is not high

enough to initiate burning, of course. Instead, if you swallow some glucose, enzymes in your cells will lower the barrier of activation energy, allowing the sugar to be oxidized in a series of steps.

Stepwise Energy Harvest via NAD+ and the Electron Transport Chain

If energy is released from a fuel all at once, it cannot be harnessed efficiently for constructive work. For example, if a gasoline tank explodes, it cannot drive a car very far. Cellular respiration does not oxidize glucose (or any other organic fuel) in a single explosive step either. Rather, glucose is broken down in a series of steps, each one catalyzed by an enzyme. At key steps, electrons are stripped from the glucose. As is often the case in oxidation reactions, each electron travels with a proton—thus, as a hydrogen atom. The hydrogen atoms are not transferred directly to O2, but instead are usually passed first to an electron carrier, a coenzyme called nicotinamide adenine dinucleotide, a derivative of the vitamin niacin. This coenzyme is well suited as an electron carrier because it can cycle easily between its oxidized form, ****NAD plus**, and its reduced form, **NADH**. As an electron acceptor, NAD+ functions as an oxidizing agent during respiration.

How does NAD+ trap electrons from glucose and the other organic molecules in food? Enzymes called dehydrogenases remove a pair of hydrogen atoms (2 electrons and 2 protons) from the substrate (glucose, in the preceding example), thereby oxidizing it. The enzyme delivers the 2 electrons along with 1 proton to its coenzyme, NAD+, forming NADH **(Figure 9.3)**. The other proton is released as a hydrogen ion H+ into the surrounding solution:

A reaction is given.

H, single bond, C, single bond, O H, with single bonds below and above C, is added to N A D plus. In the presence of dehydrogenase, it yields C, double bond O, where C has single bonds below and above, plus N A D H plus H plus.

Figure 9.3

NAD+ as an electron shuttle.

N A D plus as an electron shuttle.

Reversible reaction with enzyme dehydrogenase: to the right, reduction of N A D plus; to the left, oxidation of N A D H. Reaction is N A D plus, plus 2 H (from food) equilibrium arrow to N A D H plus H plus. The 2 H (from food) represents 2 electrons plus 2 H plus from food. One H plus, plus 2 e minus reduces N A D plus to N A D H, leaving one H plus. The structure of N A D plus is two nucleotides joined at their two phosphate groups. Nicotinamide consists of an amino group attached to a carbonyl group attached to a nitrogen-containing ring that has one more hydrogen attached in the reduced form (N A D H) than it has in the oxidized form (N A D plus).

The full name for NAD+, nicotinamide adenine dinucleotide, describes its structure—the molecule consists of two nucleotides joined together at their phosphate groups (shown in yellow). (Nicotinamide is a nitrogenous base, although not one that is present in DNA or RNA; see Figure 5.23.) The enzymatic transfer of 2 electrons and 1 proton (H+) from an organic molecule in food to NAD+reduces the NAD+to NADH: Most of the electrons removed from food are transferred initially to NAD+, forming NADH.

VISUAL SKILLS *Describe the structural differences between the oxidized form and the reduced form of nicotinamide.*

*For suggested answer, see*Appendix A.**

By receiving 2 negatively charged electrons but only 1 positively charged proton, the nicotinamide portion of NAD+ has its charge neutralized when NAD+ is reduced to NADH. The name NADH shows the hydrogen that has been received in the reaction. NAD+ is the most versatile electron acceptor in cellular respiration and functions in several of the redox steps during the breakdown of glucose.

Electrons lose very little of their potential energy when they are transferred from glucose to NAD+. Each NADH molecule formed during respiration represents stored energy that can be tapped to make ATP when the electrons complete their “fall” down an energy gradient from NADH to O2.

How do electrons that are extracted from glucose and stored as potential energy in NADH finally reach oxygen? It will help to compare the redox chemistry of cellular respiration to a much simpler reaction: the reaction between hydrogen and oxygen to form water **(Figure 9.4a)**. Mix H2 and O2, provide a spark for activation energy, and the gases combine explosively. In fact, combustion of liquid H2 and O2 is harnessed to help power the rocket engines that boost satellites into orbit and launch spacecraft. The explosion represents a release of energy as the electrons of hydrogen “fall” closer to the electronegative oxygen atoms. Cellular respiration also brings hydrogen and oxygen together to form water, but there are two important differences. First, in cellular respiration, the hydrogen that reacts with oxygen is derived from organic molecules rather than H2. Second, instead of occurring in one explosive reaction, respiration uses an electron transport chain to break the fall of electrons to oxygen into several energy-releasing steps **(Figure 9.4b)**. An **electron transport chain** consists of a number of molecules, mostly proteins, built into the inner membrane of the mitochondria of eukaryotic cells (and the plasma membrane of respiring prokaryotes). Electrons removed from glucose are shuttled by NADH to the “top,” higher-energy end of the chain. At the “bottom,” lower-energy end, O2 captures these electrons along with hydrogen nuclei H+, forming water. (Anaerobically respiring prokaryotes have an electron acceptor at the end of the chain that is different from O2.)

Figure 9.4

An introduction to electron transport chains.

An introduction to the electron transport chains.

Part a. Uncontrolled reaction. The one-step exergonic reaction of hydrogen with oxygen to form water releases a large amount of energy in the form of heat and light: an explosion. 2 H plus one-half O 2 yields one H 2 O. Free Energy (G) increases in opposition to the one-step reaction, resulting in an explosive release of heat and light energy.

Part b. Cellular respiration. The same reaction occurs in stages: An electron transport chain breaks the fall of electrons in this reaction into smaller steps and stores some of the released energy in a form that can be used to make ATP. (The rest of the energy is released as heat.) Free energy (G) increases in opposition to the stepwise progress of the reaction H 2 plus one-half O 2 goes to H 2 0, controlling the release of energy for the synthesis of A T P. 2 electrons coming from food via N A D H are passed through a series of controlled steps in the electron transport chain. At the bottom of the chain, the 2 electrons and 2 H plus combine with one-half 0 2 to form water. In the process, A T P is synthesized.

Electron transfer from NADH to oxygen is an exergonic reaction with a free-energy change of −53 kcal/mol (−222 kJ/mol). Instead of this energy being released and wasted in a single explosive step, electrons cascade down the chain from one carrier molecule to the next in a series of redox reactions, losing a small amount of energy with each step until they finally reach oxygen, the terminal electron acceptor, which has a very great affinity for electrons. Each “downhill” carrier has a greater affinity for electrons than, and is thus capable of accepting electrons from (oxidizing), its “uphill” neighbor, with O2 at the bottom of the chain. Therefore, the electrons transferred from glucose to NAD+, reducing it to NADH, fall down an energy gradient in the electron transport chain to a far more stable location in an electronegative oxygen atom from O2. Put another way, O2 pulls electrons down the chain in an energy-yielding tumble analogous to gravity pulling objects downhill.

In summary, during cellular respiration, most electrons travel the following “downhill” route: glucose→NADH→electron transport chain→oxygen. Later in this chapter, you will learn more about how the cell uses the energy released from this exergonic electron fall to regenerate its supply of ATP. For now, having covered the basic redox mechanisms of cellular respiration, let’s look at the entire process by which energy is harvested from organic fuels.

The Stages of Cellular Respiration: *A Preview*

The harvesting of energy from glucose by cellular respiration is a cumulative function of three metabolic stages. We list them here along with a color-coding scheme we will use throughout the chapter to help you keep track of the big picture:

Three stages of cellular respiration.

One. Glycolysis (color-coded blue throughout the chapter).

Two. Pyruvate oxidation (light orange) and the citric acid cycle (dark orange). Three.

Oxidative phosphorylation: electron transport and chemiosmosis (purple).

Biochemists usually reserve the term *cellular respiration* for stages 2 and 3 together. In this text, however, we include glycolysis as a part of cellular respiration because most respiring cells deriving energy from glucose use glycolysis to produce the starting material for the citric acid cycle.

As diagrammed in **Figure 9.5**, glycolysis and then pyruvate oxidation and the citric acid cycle are the catabolic pathways that break down glucose and other organic fuels. **Glycolysis**, which occurs in the cytosol, begins the degradation process by breaking glucose into two molecules of a compound called pyruvate. In eukaryotes, pyruvate enters the mitochondrion and is oxidized to a compound called acetyl CoA, which enters the **citric acid cycle**. There, the breakdown of glucose to carbon dioxide is completed. (In prokaryotes, these processes take place in the cytosol.) Thus, the carbon dioxide produced by respiration represents fragments of oxidized organic molecules.

Figure 9.5

An overview of cellular respiration.

An overview of cellular respiration.

Glycolysis occurs in the cytosol. In this process, glucose yields pyruvate. A T P is produced with substrate-level phosphorylation. Electrons are carried via N A D H into the mitochondrion. The pyruvate goes into the mitochondrion and pyruvate oxidation occurs with acetyl Co A. That goes into the citric acid cycle, which produces A T P with substrate-level phosphorylation. Electrons from the cycle are carried via N A D H and F A D H 2. These electrons and the electrons from glycolysis go into oxidative phosphorylation (electron transport and chemiosmosis). This produces A T P with oxidative phosphorylation.

During glycolysis, each glucose molecule is broken down into two molecules of pyruvate. In eukaryotic cells, as shown here, the pyruvate enters the mitochondrion. There it is oxidized to acetyl CoA, which will be further oxidized to CO_2 in the citric acid cycle. The electron carriers NADH and $FADH_2$ transfer electrons derived from glucose to electron transport chains. During oxidative phosphorylation, electron transport chains convert the chemical energy to a form used for ATP synthesis in the process called chemiosmosis. (During earlier steps of cellular respiration, a few molecules of ATP are synthesized in a process called substrate-level phosphorylation.) To visualize these processes in their cellular context, see Figure 6.32b.

Animation

Overview of Cellular Respiration

image

Play

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Some of the steps of glycolysis and the citric acid cycle are redox reactions in which dehydrogenases transfer electrons from substrates to NAD+ or the related electron carrier FAD, forming NADH or FADH2. (You'll learn more about FAD and FADH2 later.) In the third stage of respiration, the electron transport chain accepts electrons from NADH or FADH2 generated during the first two stages and passes these electrons down the chain. At the end of the chain, the electrons are combined with molecular oxygen O2 and hydrogen ions H+, forming water (see Figure 9.4b). The energy released at each step of the chain is stored in a form the mitochondrion (or prokaryotic cell) can use to make ATP from ADP. This mode of ATP synthesis is called **oxidative phosphorylation** because it is powered by the redox reactions of the electron transport chain.

In eukaryotic cells, the inner membrane of the mitochondrion is the site of electron transport and another process called *chemiosmosis*, together making up oxidative phosphorylation. (In prokaryotes, these processes take place in the plasma membrane.) Oxidative phosphorylation accounts for almost 90% of the ATP generated by respiration. A smaller amount of ATP is formed directly in a few reactions of glycolysis and the citric acid cycle by a mechanism called **substrate-level phosphorylation(Figure 9.6)**. This mode of ATP synthesis occurs when an enzyme transfers a phosphate group from a substrate molecule to ADP, rather than adding an inorganic phosphate to ADP as in oxidative phosphorylation. "Substrate molecule" here refers to an organic molecule generated as an intermediate during the catabolism of glucose. You'll see examples of substrate-level phosphorylation later in the chapter, in both glycolysis and the citric acid cycle.

Figure 9.6

Substrate-level phosphorylation.

Substrate with attached phosphate group binds to the active site of an enzyme, which transfers phosphate to ADP, forming ATP. Substrate is then released by the enzyme as a product.

Some ATP is made by direct transfer of a phosphate group from an organic substrate to ADP by an enzyme. (For examples in glycolysis, see Figure 9.8, steps 7 and 10.)

MAKE CONNECTIONS *Review*Figure 8.9*. In the reaction shown above, is the potential energy higher for the reactants or for the products? Explain.*

*For suggested answer, see*Appendix A*.*

You can think of the whole process this way: When you withdraw a relatively large sum of money from an ATM, it is not delivered to you in a single bill of a large denomination. Instead, the machine dispenses a number of smaller-denomination bills that you can spend more easily. This is analogous to ATP production during cellular respiration. For each molecule of glucose degraded to CO_2 and H_2O by respiration, the cell makes up to about 32 molecules of ATP, each with 7.3 kcal/mol of free energy. Respiration cashes in the large denomination of energy banked in a single molecule of glucose (686 kcal/mol under standard conditions) for the small change of many molecules of ATP, which is more practical for the cell to spend on its work.

This preview has introduced you to how glycolysis, the citric acid cycle, and oxidative phosphorylation fit into the process of cellular respiration so you can keep the big picture in mind as you take a closer look at each of these three stages of respiration. As you read about the chemical reactions, remember that each reaction is catalyzed by a specific enzyme, some of which are shown in Figure 6.32b.

Concept 9.2: Glycolysis harvests chemical energy by oxidizing glucose to pyruvate

The word *glycolysis* means “sugar splitting,” and that is exactly what happens during this pathway. Glucose, a six-carbon sugar, is split into two three-carbon sugars. These smaller sugars are then oxidized and their remaining atoms rearranged to form two molecules of pyruvate. (Pyruvate is the ionized form of pyruvic acid.)

As summarized in **Figure 9.7**, glycolysis can be divided into two phases: the energy investment phase and the energy payoff phase. During the energy investment phase, the cell actually spends ATP. This investment is repaid with interest during the energy payoff phase, when ATP is produced by substrate-level phosphorylation and NAD^+ is reduced to NADH by electrons released from the oxidation of glucose. The net energy yield from glycolysis, per glucose molecule, is 2 ATP plus 2 NADH. The ten steps of the glycolytic pathway are shown in **Figure 9.8**.

Figure 9.7

The inputs and outputs of glycolysis.

The inputs and outputs of glycolysis.

Overview: Glycolysis takes place in the cytoplasm and produces a small amount of ATP. The outputs from glycolysis will enter pyruvate oxidation, the citric acid cycle, and oxidative phosphorylation inside the mitochondrion. Inputs and output from glycolysis are summarized as follows: Energy Investment Phase: 2 A T P goes to 2 A D P plus 2 inorganic phosphates. Glucose enters the pathway. Energy payoff phase: 4 A D P plus 4 inorganic phosphate goes to 4 A T P. 2 N A D plus, plus 4 e minus plus 4 H plus goes to 2 N A D H plus 2 H plus. Glucose is split into 2 pyruvates plus 2 H 2 0. Net: Glucose goes to 2 pyruvates plus 2 H 2 0. 4 ATP formed minus 2 A T P used equals to 2 A T P. 2 N A D plus, plus 4 e minus plus 4 H plus goes to 2 N A D H plus 2 H plus.

Animation

Glycolysis

image

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Figure 9.8

The steps of glycolysis.

The steps of glycolysis.

Hexokinase transfers a phosphate group from ATP to glucose, making it more chemically reactive. The charge on the phosphate also traps the sugar in the cell. The structure of glucose is a hexagon (a six-carbon ring with carbons 1-5 forming 5 corners of the ring and O attached to carbon 5 at the sixth corner). O H and H are attached to carbons 1-4. C H 2 O H is attached to carbon 5, as is O in the sixth corner of the ring. One A T P is used (going to A D P) with the enzyme hexokinase to form glucose 6-phosphate, which has the same structure as glucose, except that there is a phosphate group (from A T P) attached above carbon 5, forming C H 2 O attached to a phosphate group. Glucose 6-phosphate is converted to fructose 6-phosphate. The enzyme involved in this conversion is phosphoglucoisomerase. Glucose 6-phosphate (structure described in a) is converted to fructose 6-phosphate with the enzyme phosphoglucoisomerase. The structure of fructose 6-phosphate is a pentagon, with carbon at all unlabeled corners. Attached to carbon 1 are C H 2 O H and H. Attached to carbons 2 and 3 are O H and H. Attached to carbon 4 are H and C H 2 O H attached to a phosphate group. At the point of the pentagon is an O. Phosphofructokinase transfers a phosphate group from A T P to the opposite end of the sugar, investing a second molecule of A T P. This is a key step for the

regulation of glycolysis. Fructose 6-phosphate is converted to fructose 1,6-bisphosphate as the enzyme phosphofructokinase transfers a phosphate group from ATP. Fructose 1,6-bisphosphate has the same structure as fructose 6 phosphate (described in b), with the addition of a phosphate group (from A T P) attached above carbon 1, forming C H 2 O attached to a phosphate group. Aldolase cleaves the sugar molecule into two different three-carbon sugars. Aldolase cleaves fructose 1,6-bisphosphate (structure described in c) into glyceraldehyde 3-phosphate (G3P) and dihydroxyacetone phosphate (D H A P). Structure of glyceraldehyde 3-phosphate is a 3-carbon chain with an H and an O (double bond) attached to carbon 1, an H O H attached to carbon 2 and an H 2 0, single bond, P attached to carbon 3. Structure of dihydroxyacetone phosphate is a 3 carbon chain with H 2 0 attached to a phosphate group attached to carbon 1, an O (double bond) attached to carbon 2 and an H 2 O H attached to carbon 3. Conversion between D H A P and G 3 P: This reaction never reaches equilibrium; G 3 P is used in the next step as fast as it forms. Isomerase catalyzes the reversible reaction between glyceraldehyde 3-phosphate and dihydroxyacetone phosphate (structures in d). Continued on next page.

Two sequential reactions: (1) 2 N A D plus are reduced to 2 N A D H plus 2 H plus. (2) G 3 P is converted to 1,3 bisphosphoglycerate.

Two sequential reactions: (1) G3P is oxidized by the transfer of electrons to N A D plus, forming N A D H. (2) Using energy from this exergonic redox reaction, a phosphate group is attached to the oxidized substrate, making a high-energy product.

In a reaction catalyzed by triose phosphate dehydrogenase and with an input of 2 inorganic phosphate molecules, glyceraldehyde 3-phosphate (G 3 P) is converted to 1,3 bisphosphoglycerate. The structure is a 3-carbon chain P, single bond, O and O (double bond) attached to carbon 1, H O H attached to carbon 2, and H 2 0 attached to a phosphate group attached to carbon 3. The phosphate group is transferred to ADP (substrate-level phosphorylation) in an exergonic reaction. The carbonyl group of G3P has been oxidized to the carboxyl group (C O O minus) of an organic acid (3-phosphoglycerate). 2 A T P are formed from 2 A D P through substrate-level phosphorylation. In a reaction catalyzed by phosphoglycerokinase, 1,3-Bisphosphoglycerate (structure described in f) is converted to 3-phosphoglycerate. Structure is a 3 carbon chain with an O minus and another O (double bond) attached to carbon 1, an H O H attached to carbon 2, and an H 2 0 attached to a phosphate group attached to carbon 3. This enzyme relocates the remaining phosphate group. 3-phosphoglycerate is converted to 2-phosphoglycerate in a reaction catalyzed by phosphoglyceromutase. 3-phosphoglycerate (structure described in g) is converted to 2-phosphoglycerate. Structure is the same as 3-phosphoglycerate, except the enzyme phosphoglyceromutase relocates the phosphate group from the location in 3-phosphoglycerate to the second carbon in 2-phosphoglycerate so that H and O with a phosphate group are now attached at carbon 2. Enolase causes a double bond to form in the substrate by extracting a water molecule, yielding phosphoenolpyruvate (P E P), a compound with a very high potential energy. 2-phosphoglycerate is converted to P E P when enolase extracts a water molecule,

causing a double bond to form in P E P. Structure is the same as 2-phosphoglycerate, except there is a double bond between carbon 3 and C H 2 (was C H 2 O H) and there is no longer an H on carbon 2. The phosphate group is transferred from P E P to A D P (a second example of substrate-level phosphorylation), forming pyruvate. PEP (structure described in i) is converted to pyruvate in a reaction catalyzed by pyruvate kinase. 2 A T P are formed from 2 A D P during the conversion by substrate level phosphorylation. Structure of pyruvate is the same as P E P, except the phosphate group has been removed from carbon 2 and carbon 3 is now C H 3.

Glycolysis, a source of ATP and NADH, takes place in the cytosol. Two of the enzymes (in **STEPS 1** and **3**) are shown in Figure 6.32b.

WHAT IF? *What would happen if you removed the dihydroxyacetone phosphate generated in step 4 as fast as it was produced?*

*For suggested answer, see*Appendix A.**

BioFlix® Animation

Glycolysis

image

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All of the carbon originally present in glucose is accounted for in the two molecules of pyruvate; no carbon is released as CO2 during glycolysis. Glycolysis occurs whether or not O2 is present. However, if O2*is* present, the chemical energy stored in pyruvate and NADH can be extracted by pyruvate oxidation, the citric acid cycle, and oxidative phosphorylation.

Concept 9.3: After pyruvate is oxidized, the citric acid cycle completes the energy-yielding

oxidation of organic molecules

Glycolysis releases less than a quarter of the chemical energy in glucose that can be harvested by cells; most of the energy remains stockpiled in the two molecules of pyruvate. When O_2 is present, the pyruvate in eukaryotic cells enters a mitochondrion, where the oxidation of glucose is completed. In aerobically respiring prokaryotic cells, this process occurs in the cytosol. (Later in the chapter, we'll examine other fates for pyruvate—for example, when O_2 is unavailable or in a prokaryote that is unable to use O_2.)

Oxidation of Pyruvate to Acetyl CoA

Upon entering the mitochondrion via active transport, pyruvate is first converted to a compound called acetyl coenzyme A, or **acetyl CoA(Figure 9.9)**. This step, linking glycolysis and the citric acid cycle, is carried out by a multienzyme complex that catalyzes three reactions: **STEP 1** Pyruvate's carboxyl group (—COO−), already somewhat oxidized and thus carrying little chemical energy, is now fully oxidized and given off as a molecule of CO_2. This is the first step in which CO_2 is released during respiration. **STEP 2** Next, the remaining two-carbon fragment is oxidized and the electrons transferred to NAD^+, storing energy in the form of NADH. **STEP 3** Finally, coenzyme A (CoA), a sulfur-containing compound derived from a B vitamin, is attached via its sulfur atom to the two-carbon intermediate, forming acetyl CoA. Acetyl CoA has a high potential energy, which is used to transfer the acetyl group to a molecule in the citric acid cycle, a reaction that is therefore highly exergonic.

Figure 9.9

Oxidation of pyruvate to acetyl CoA, the step before the citric acid cycle.

Oxidation of pyruvate to acetyl Co A.

Pyruvate in the cytosol. It moves into the mitochondrion via a transport protein. Pyruvate dehydrogenase releases C O 2. N A D plus reacts with the pyruvate dehydrogenase and becomes N A D H plus and H plus. Coenzyme A reacts with pyruvate dehydrogenase to produce acetyl Co A.

Pyruvate enters the mitochondrion through a transport protein and is processed by a complex of several enzymes known as pyruvate dehydrogenase (shown in the computer-generated image based on cryo-EMs). This complex catalyzes the three numbered steps, which are described in the text. The CO_2 molecule will diffuse out of the cell. The NADH will be used in oxidative phosphorylation. The acetyl group of acetyl CoA will enter the citric acid cycle. (Coenzyme A is abbreviated S-CoA when it is attached to a molecule, emphasizing its sulfur atom, S.)

BioFlix® Animation

Acetyl CoA

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The Citric Acid Cycle

The citric acid cycle functions as a metabolic furnace that further oxidizes organic fuel derived from pyruvate. **Figure 9.10** summarizes the inputs and outputs as pyruvate is broken down to three CO_2 molecules, including the molecule of CO_2 released during the conversion of pyruvate to acetyl CoA. The cycle generates 1 ATP per turn by substrate-level phosphorylation, but most of the chemical energy is transferred to NAD+ and FAD during the redox reactions. The reduced coenzymes, NADH and $FADH_2$, shuttle their cargo of high-energy electrons into the electron transport chain. The citric acid cycle is also called the tricarboxylic acid cycle or the Krebs cycle, the latter honoring Hans Krebs. Krebs was the German-British scientist largely responsible for working out the pathway in the 1930s.

Figure 9.10

An overview of pyruvate oxidation and the citric acid cycle.

Pyruvate oxidation: pyruvate goes to Acetyl Co A, giving off C O 2; N A D plus is reduced to N A D H. Citric acid cycle: Acetyl Co A enters, outputs are: C O 2, N A D H, F A D H 2 and A T P.

Overview: Inputs to pyruvate oxidation come from glycolysis. Outputs from pyruvate oxidation will enter the citric acid cycle, which is followed by oxidative phosphorylation. All steps except glycolysis, which occurs in the cytosol, take place inside the mitochondrion.

Pyruvate oxidation: 3-carbon molecule of pyruvate (from glycolysis; 2 per glucose) goes to a 2-carbon molecule attached to coenzyme A, forming acetyl CoA. In the conversion, CO_2 is given off and N A D plus is reduced to N A D H.

Citric Acid Cycle: Acetyl Co A from the previous step enters, and Co A is released. Per molecule entering the cycle, 2 molecules of C O 2 are released, 3 N A D plus are reduced to N A D H plus 3 H plus, A T P is formed

from A D P plus inorganic phosphate, and F A D is reduced to F A D H 2.

The inputs and outputs per pyruvate molecule are shown with a focus on the carbon atoms involved. To calculate on a per-glucose basis, multiply by 2 because each glucose molecule is split during glycolysis into two pyruvate molecules.

Animation

The Citric Acid Cycle

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Now let's look at the citric acid cycle in more detail. The cycle has eight steps, each catalyzed by a specific enzyme. You can see in **Figure 9.11** that for each turn of the citric acid cycle, two carbons (red) enter in the relatively reduced form of an acetyl group (**STEP 1**), and two different carbons (blue) leave in the completely oxidized form of CO2 molecules (**STEPS 3** and **4**). The acetyl group of acetyl CoA joins the cycle by combining with the compound oxaloacetate, forming citrate (**STEP 1**). Citrate is the ionized form of citric acid, for which the cycle is named. The next seven steps decompose the citrate back to oxaloacetate. It is this regeneration of oxaloacetate that makes the process a *cycle*.

Figure 9.11

A closer look at the citric acid cycle.

A closer look at the citric acid cycle.

Acetyl Co A (from oxidation of pyruvate) adds its two-carbon acetyl group to oxaloacetate, producing citrate. Acetyl Co-A is a 2-carbon chain with both carbons shown in red so that they can be tracked throughout the cycle. Attached to carbon 1 are S C o A and an O (double bond). Attached to carbon 2 is H 3. The two carbons from Acetyl Co A are added to oxaloacetate. The structure of oxaloacetate is a three-carbon chain with an O (double bond) and an ionized carboxylic acid (C O O minus) attached to carbon 1, H 2 attached to carbon 2, and carbon 3 is part of another ionized carboxylic acid. Both carbons in the carboxylic acids are blue, indicating that they will later be the carbons released as C O 2. The addition of acetyl Co A to oxaloacetate produces citrate, a chain of five carbons with carbons 1 and 2 coming from acetyl Co A (carbon one is in a carboxylic acid and carbon 2 has H 2 attached); two more carboxylic acids attached to carbons 3 and 5 that

will be released as C O 2 in later steps, and an O H on carbon 3 as well as an H 2 on carbon 4. Citrate is converted to its isomer, isocitrate, by the removal of one water molecule and the addition of another. Isocitrate is oxidized, reducing N A D plus to N A D H. Then the resulting compound loses a C O 2 molecule. Isocitrate oxidized to alpha-ketoglutarate. The C O 2 released came from the carbon in the carboxylic acid attached off carbon 3 in step b. The structure of alpha-ketoglutarate is a 5 carbon chain, with the first two carbons shown in red to indicate that they came from acetyl Co A. Carbon 1 is part of a carboxylic acid; carbon 2 and 3 have an H 2, carbon 4 has a double bond to O, and carbon 5 is a carboxylic acid, shown in blue to indicate it will later be released as C O 2. Another C O 2 is lost, and the resulting compound is oxidized, reducing N A D plus to N A D H. The remaining molecule is then attached to coenzyme A by an unstable bond. Alpha-ketoglutarate goes to Succinyl Co A. Structure of Succinyl Co A is the same as alpha-ketoglutarate, except that the carboxylic acid at carbon 5 is gone (the C released in C O 2) and attached in its place is S-Co A. Co A is displaced by a phosphate group, which is transferred to G D P, forming G T P, a molecule with functions similar to ATP. G T P can also be used, as shown, to generate A T P. Succinyl Co A goes to succinate; structure is a 4 carbon chain with carboxylic acids on carbons 1 and 4 and H 2 S on carbons 2 and 3. Two hydrogens are transferred to F A D, forming F A D H 2 and oxidizing succinate. Succinate goes to fumarate; structure like succinate except there is a double bond between carbons 2 and 3. Addition of a water molecule rearranges bonds in the substrate. Fumarate goes to malate; structure like fumarate except no double bond; O H and H attached to carbon 2 and H 2 attached to carbon 3.

In the chemical structures, red type traces the fate of the two carbon atoms that enter the cycle via acetyl CoA (**STEP 1**), and blue type indicates the two carbons that exit the cycle as CO_2 in **STEPS 3** and **4**. (The red type goes only through **STEP 5** because the succinate molecule is symmetrical; the two ends cannot be distinguished from each other.) Notice that the carbon atoms that enter the cycle from acetyl CoA do not leave the cycle in the same turn. They remain in the cycle, occupying a different location in the molecules on their next turn, after another acetyl group is added. Therefore, the oxaloacetate regenerated at **STEP 8** is made up of different carbon atoms each time around. Carboxylic acids are represented in their ionized forms, as —COO−, because the ionized forms prevail at the pH within the mitochondrion. In eukaryotic cells, all the citric acid cycle enzymes are located in the mitochondrial matrix except for the enzyme that catalyzes **STEP 6**, which resides in the inner mitochondrial membrane. (The enzyme that catalyzes **STEP 3**, isocitrate dehydrogenase, is shown in Figure 6. 32b.)

BioFlix® Animation

The Citric Acid Cycle

image

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image

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Referring to Figure 9.11, we can tally the energy-rich molecules produced by the citric acid cycle. For each acetyl group entering the cycle, 3 NAD+ are reduced to NADH (**STEPS 3**, **4**, and **8**). In **STEP 6**, electrons are transferred not to NAD+, but to FAD, which accepts 2 electrons and 2 protons to become FADH2. In many animal tissue cells, the reaction in **STEP 5** produces a guanosine triphosphate (GTP) molecule by substrate-level phosphorylation. GTP is a molecule similar to ATP in its structure and cellular function. This GTP may be used to make an ATP molecule (as shown) or directly power work in the cell. In tFhe cells of plants, bacteria, and some animal tissues, **STEP 5** forms an ATP molecule directly by substrate-level phosphorylation. The output from **STEP 5** represents the only ATP generated during the citric acid cycle. Recall that each glucose gives rise to two molecules of acetyl CoA that enter the cycle. Because the numbers noted earlier are obtained from a single acetyl group entering the pathway, the total yield per glucose from the citric acid cycle turns out to be doubled, or 6 NADH , 2 FADH2, and the equivalent of 2 ATP.

Most of the ATP produced by respiration is generated later, from oxidative phosphorylation, when the NADH and FADH2 produced by the citric acid cycle and earlier steps relay the electrons extracted from food to the electron transport chain. In the process, they supply the necessary energy for the phosphorylation of ADP to ATP. We will explore this process in the next section.

Concept 9.4: During oxidative phosphorylation, chemiosmosis couples electron transport to ATP synthesis

Our main objective in this chapter is to learn how cells harvest the energy of glucose and other nutrients in food to make ATP. But the metabolic components of respiration we have examined so far, glycolysis and the citric acid cycle, produce only 4 ATP molecules per glucose molecule, all by substrate-level phosphorylation: 2 net ATP from glycolysis and 2 ATP from the citric acid cycle. At this point, molecules of NADH (and FADH2) account for most of the energy extracted from each glucose molecule. These electron escorts link glycolysis and the citric acid cycle to the machinery of oxidative phosphorylation, which uses energy released

by the electron transport chain to power ATP synthesis. In this section, you will learn first how the electron transport chain works and then how electron flow down the chain is coupled to ATP synthesis.

The Pathway of Electron Transport

The electron transport chain is a collection of molecules embedded in the inner membrane of the mitochondrion in eukaryotic cells. (In prokaryotes, these molecules reside in the plasma membrane.) The folding of the inner membrane to form cristae increases its surface area, providing space for thousands of copies of each component of the electron transport chain in a mitochondrion. Structure fits function: The infolded membrane with its concentration of electron carrier molecules is well-suited for the series of sequential redox reactions that take place along the electron transport chain. Most components of the chain are proteins, which exist in multiprotein complexes numbered I through IV. Tightly bound to these proteins are *prosthetic groups*, nonprotein components such as cofactors and coenzymes essential for the catalytic functions of certain enzymes.

Figure 9.12 shows the sequence of electron carriers in the electron transport chain and the drop in free energy as electrons travel down the chain. During this electron transport, electron carriers alternate between reduced and oxidized states as they accept and then donate electrons. Each component of the chain becomes reduced when it accepts electrons from its “uphill” neighbor, which is a better electron donor. It then returns to its oxidized form as it passes electrons to its “downhill” neighbor, which is a better electron acceptor.

Figure 9.12

Free-energy change during electron transport.

Graphical representation of free energy change during electron transport. Y-axis is free energy (G) relative to O 2 (k cal per mol). Prosthetic groups in electron transport chain show a downhill slope toward O 2.

Overview: Steps in cellular respiration: glycolysis, pyruvate oxidation, citric acid cycle, and oxidative phosphorylation (produces A T P). All steps except glycolysis take place in the mitochondrion. 2 electrons from N A D H enter the electron transport chain, oxidizing N A D H to N A D plus. The electrons are passed downhill through three protein complexes, with several prosthetic groups bound in each. Flow of electrons from N A D H (least electronegative): Complex I: F M N to F E S. Between Complex 1 and 3 electrons pass to Q. In Complex 3 electrons flow from Cyt b to F E S to Cyt C sub 1. Between Complex 3 and 4 electrons pass to Cyt c. In Complex 4 electrons pass to Cyt a and then Cyt a sub 3. 2 electrons then join one-half O2 and 2 H plus (most electronegative) to form H 2 0. Complexes I-IV each consist of multiple proteins with electron carriers. 2 electrons from F A D H 2 can also join the chain from Complex 2, passing through F E S and then

joining Q between Complexes 1 and 3, from which point they will follow the same path as electrons from N A D H.

Electron transport chain: Electrons (from NADH or F A D H 2) move from a less electronegative electron carrier (one with a lower affinity for electrons) to a more electronegative electron carrier down the chain, releasing free energy. The last electron carrier (Cyt a3) passes its electrons to oxygen, which is very electronegative.

The overall energy drop (ΔG) for electrons traveling from NADH to oxygen is 53 kcal/mol, but this “fall” is broken up into a series of smaller steps by the electron transport chain. (An oxygen atom is represented here as 12O2 to show that O2 is reduced, not individual oxygen atoms.) To view these proteins in their cellular context, see Figure 6.32b.

VISUAL SKILLS *Compare the position of the electrons in NADH (see*Figure 9.3*)* at the top of the chain with that in*H2O*, at the bottom. Describe why the electrons in*H2O*have less potential energy, using the term electronegativity.*

Figure 9.12 Walkthrough

Free-energy change during electron transport

Watch an explanation of this figure in the Figure 9.12 Walkthrough: Free-energy change during electron transport.

image

image

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Now let’s take a closer look at the electrons as they drop in energy level, passing through the components of the electron transport chain in Figure 9.12. We’ll first look at the passage of electrons through complex I in some detail as an illustration of the general principles involved in electron transport. Electrons acquired from glucose by NAD+ during glycolysis and the citric acid cycle are transferred from NADH to the first molecule of the electron transport chain in complex I. This molecule is a flavoprotein, so named because it has a prosthetic group called flavin mononucleotide (FMN). In the next redox reaction, the flavoprotein returns to its oxidized form as it passes electrons to an iron-sulfur protein (Fe·S in complex I), one of a family of proteins with both iron and sulfur tightly bound. The iron-sulfur protein then passes the electrons to a compound called ubiquinone (Q in Figure 9.12). This electron carrier is a small hydrophobic molecule, the only member of the electron transport chain that is not a protein. Ubiquinone is individually mobile

within the membrane rather than residing in a particular complex. (Another name for ubiquinone is coenzyme Q, or CoQ; you may have seen it sold as a nutritional supplement.)

Most of the remaining electron carriers between ubiquinone and oxygen are proteins called **cytochromes**. Their prosthetic group, called a heme group, has an iron atom that accepts and donates electrons. (The heme group in a cytochrome is similar to the heme group in hemoglobin, the protein of red blood cells, except that the iron in hemoglobin carries oxygen, not electrons.) The electron transport chain has several types of cytochromes, each named "cyt" with a letter and number to distinguish it as a different protein with a slightly different electron-carrying heme group. The last cytochrome of the chain, cyt a3, passes its electrons to oxygen (in O_2), which is *very* electronegative. Each O also picks up a pair of hydrogen ions (protons) from the aqueous solution, neutralizing the −2 charge of the added electrons and forming water.

Another source of electrons for the electron transport chain is $FADH_2$, the other reduced product of the citric acid cycle. Notice in Figure 9.12 that $FADH_2$ adds its electrons from within complex II, at a lower energy level than NADH does. Consequently, although NADH and $FADH_2$ each donate an equivalent number of electrons (2) for oxygen reduction, the electron transport chain provides about one-third less energy for ATP synthesis when the electron donor is $FADH_2$ rather than NADH. We'll see why in the next section.

The electron transport chain makes no ATP directly. Instead, it eases the fall of electrons from food to oxygen, breaking a large free-energy drop into a series of smaller steps that release energy in manageable amounts, step by step. How does the mitochondrion (or the plasma membrane in prokaryotes) couple this electron transport and energy release to ATP synthesis? The answer is a mechanism called chemiosmosis.

Chemiosmosis: The Energy-Coupling Mechanism

Populating the inner membrane of the mitochondrion or the prokaryotic plasma membrane are many copies of a protein complex called **ATP synthase**, the enzyme that makes ATP from ADP and inorganic phosphate **(Figure 9.13)**. ATP synthase works like an ion pump running in reverse. Ion pumps usually use ATP as an energy source to transport ions against their gradients. Enzymes can catalyze a reaction in either direction, depending on the ΔG for the reaction, which is affected by the local concentrations of reactants and products (see Concepts 8.2 and 8.3). Under the conditions of cellular respiration, rather than hydrolyzing ATP to pump protons against their concentration gradient, ATP synthase uses the energy of an existing ion gradient to power ATP synthesis. The power source for ATP synthase is a difference in the concentration of H^+ (a pH difference) on opposite sides of the inner mitochondrial membrane. This process, in which energy stored in the form of a hydrogen ion gradient across a membrane is used to drive cellular work such as the synthesis of

ATP, is called **chemiosmosis** (from the Greek *osmos*, push). The word *osmosis* was previously used in discussing water transport, but here it refers to the flow of H+ across a membrane.

Figure 9.13

ATP synthase, a molecular mill.

A T P synthase in mitochondrial matrix, attached to rotor imbedded in the membrane through the internal rod and the catalytic rod; stator adjacent to the rotor open to the intermembrane space.

H plus ions flowing down their gradient enter a channel in a stator, which is anchored in the membrane. H plus ions are concentrated in the intermembrane space until they enter the stator. There is a clear concentration gradient of H plus ions across the mitochondrial membrane. H plus ions enter binding sites within a rotor, changing the shape of each subunit so that the rotor spins within the membrane. Each H plus ion makes one complete turn before leaving the rotor and passing through a second channel in the stator into the mitochondrial matrix. The spinning of the rotor causes an internal rod to spin as well. This rod extends like a stalk into the knob below it, which is held stationary by part of the stator. Turning the rod activates catalytic sites in the knob that produce A T P from A D P and P sub i.

Multiple ATP synthases reside in eukaryotic mitochondrial and chloroplast membranes and in prokaryotic plasma membranes. (See Figure 6.32b and c.)

BioFlix® Animation

ATP Synthase

image

image

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Animation

Rotating ATP Synthase

image

image

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From studying the structure of ATP synthase, scientists have learned how the flow of H+ through this enzyme powers ATP generation. ATP synthase is a multisubunit complex with four main parts, each made

up of multiple polypeptides (see Figure 9.13). Protons move one by one into binding sites on the rotor, causing it to spin in a way that catalyzes ATP production from ADP and Ⓟi. The flow of protons thus behaves somewhat like a rushing stream that turns a waterwheel. ATP synthase is the smallest molecular rotary motor known in nature.

How does the inner mitochondrial membrane (or the prokaryotic plasma membrane) generate and maintain the H+ gradient that drives ATP synthesis by the ATP synthase protein complex? Establishing the H+ gradient is a major function of the electron transport chain, which is shown in its mitochondrial location in **Figure 9.14**. The chain is an energy converter that uses the exergonic flow of electrons from NADH and FADH2 to pump H+ across the membrane, from the mitochondrial matrix into the intermembrane space. The H+ has a tendency to move back across the membrane, diffusing down its gradient. And the ATP synthases are the only sites that provide a route through the membrane for H+. As we described previously, the passage of H+ through ATP synthase uses the exergonic flow of H+ to drive the phosphorylation of ADP. Thus, the energy stored in an H+ gradient across a membrane couples the redox reactions of the electron transport chain to ATP synthesis.

Figure 9.14

Chemiosmosis couples the electron transport chain to ATP synthesis.

In the electron transport chain, electron transport through a series of protons embedded in the inner mitochondrial membrane causes protons to be pumped, creating an H plus gradient across the membrane.

Inner mitochondrial membrane separates mitochondrial matrix from intermembrane space. N A D H carries electrons from food to the electron transport chain, a series of protein complexes of electron carriers embedded in the inner mitochondrial membrane. The electrons are passed through Complex 1, and a proton is pumped into the intermembrane space. The electrons then pass downhill to Q, then to Complex 3, where a proton is pumped into the intermembrane space. The electrons pass through Cyt c and then to Complex 4, where a proton is pumped into the intermembrane space, and at the bottom of this last complex, electrons join 2 H plus and one-half O 2 to form H 2 O. Electrons from F A D H 2 can also join the electron transport chain at Q, from which point they will follow the same path as electrons from N A D H.

Chemiosmosis: A T P synthase is powered by the flow of H plus back across the inner mitochondrial membrane, forming A T P. Protons pumped into the intermembrane space during the electron transport chain fall back down their concentration gradient through the A T P synthase (embedded in the inner mitochondrial membrane), causing inorganic phosphate to be added to A D P, forming A T P.

STEP 1 NADH and FADH2 shuttle high-energy electrons extracted from food during glycolysis and the citric acid cycle into an electron transport chain built into the inner mitochondrial membrane. (See Figure 6.32b.) The gold arrows trace the transport of electrons, which are finally passed to a terminal acceptor (O2,

in the case of aerobic respiration) at the “downhill” end of the chain, forming water. Most of the electron carriers of the chain are grouped into four complexes (I–IV). Two mobile carriers, ubiquinone (Q) and cytochrome c (Cyt c), move rapidly, ferrying electrons between the large complexes. As the complexes shuttle electrons, they pump protons from the mitochondrial matrix into the intermembrane space. FADH2 deposits its electrons via complex II and so results in fewer protons being pumped into the intermembrane space than occurs with NADH. Chemical energy originally harvested from food is transformed into a proton-motive force, a gradient of H+ across the membrane. **STEP 2** During chemiosmosis, the protons flow back down their gradient via ATP synthase, which is built into the membrane nearby. The ATP synthase harnesses the proton-motive force to phosphorylate ADP, forming ATP. Together, electron transport and chemiosmosis make up oxidative phosphorylation.

WHAT IF? *If complex IV were nonfunctional, could chemiosmosis produce any**ATP**, and if so, how would the rate of synthesis differ?*

*For suggested answer, see*Appendix A.**

Animation

Electron Transport

image

image

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BioFlix® Animation

Electron Transport

image

image

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At this point, you may be wondering how the electron transport chain pumps hydrogen ions into the intermembrane space. Researchers have found that certain members of the electron transport chain accept and release protons H+ along with electrons. (The aqueous solutions inside and surrounding the cell are a ready source of H+.) At certain steps along the chain, electron transfers cause H+ to be taken up and released into the surrounding solution. In eukaryotic cells, the electron carriers are spatially arranged in the inner mitochondrial membrane in such a way that H+ is accepted from the mitochondrial matrix and deposited in the intermembrane space (see Figure 9.14). The H+ gradient that results is referred to as a

proton-motive force, emphasizing the capacity of the gradient to perform work. The force drives H+ back across the membrane through the H+ channels provided by ATP synthases.

In general terms, *chemiosmosis is an energy-coupling mechanism that uses energy stored in the form of an*H+*gradient across a membrane to drive cellular work*. In mitochondria, the energy for gradient formation comes from exergonic redox reactions along the electron transport chain, and ATP synthesis is the work performed. But chemiosmosis also occurs elsewhere and in other variations. Chloroplasts use chemiosmosis to generate ATP during photosynthesis; in these organelles, light (rather than chemical energy) drives both electron flow down an electron transport chain and the resulting H+ gradient formation. Prokaryotes, as already mentioned, generate H+ gradients across their plasma membranes. They then tap the proton-motive force not only to make ATP inside the cell but also to rotate their flagella and to pump nutrients and waste products across the membrane. Because of its central importance to energy conversions in prokaryotes and eukaryotes, chemiosmosis has helped unify the study of bioenergetics. Peter Mitchell was awarded the Nobel Prize in 1978 for originally proposing the chemiosmotic model.

An Accounting of ATP Production by Cellular Respiration

In the last few sections, we have looked rather closely at the key processes of cellular respiration. Now let's take a step back and remind ourselves of its overall function: harvesting the energy of glucose for ATP synthesis.

During respiration, most energy flows in this sequence: glucose→NADH→electron transport chain→protonmotive force→ATP. We can do some bookkeeping to calculate the ATP profit when cellular respiration oxidizes a molecule of glucose to six molecules of carbon dioxide. The three main departments of this metabolic enterprise are glycolysis, pyruvate oxidation and the citric acid cycle, and the electron transport chain, which drives oxidative phosphorylation. **Figure 9.15** gives a detailed accounting of the ATP yield for each glucose molecule that is oxidized. The tally adds the 4 ATP produced directly by substrate-level phosphorylation during glycolysis and the citric acid cycle to the many more molecules of ATP generated by oxidative phosphorylation. Each NADH that transfers a pair of electrons from glucose to the electron transport chain contributes enough to the proton-motive force to generate a maximum of about 3 ATP.

Figure 9.15

ATP yield per molecule of glucose at each stage of cellular respiration.

A T P yield per molecule of glucose at each stage of cellular respiration.

In the cytosol, glycolysis produces 2 A T P via substrate-level phosphorylation; 2 pyruvates from glucose which enter pyruvate oxidation (inside the mitochondrion); and 2 NADH which carry electrons to oxidative phosphorylation inside the mitochondrion. In pyruvate oxidation, 2 pyruvates (from glycolysis) are converted to Acetyl Co-A, which will go to the citric acid cycle. 2 N A D H also carry electrons to oxidative phosphorylation. In the citric acid cycle, 2 A T P are formed, ad 6 N A D H and 2 F A D H 2 carry electrons to oxidative phosphorylation. Oxidative phosphorylation produces about 26 or 28 A T P by oxidative phosphorylation, depending on which shuttles transports electrons from N A D H in the cytosol. The maximum per glucose formation of A T P is about 30 or 32 A T P.

VISUAL SKILLS *After reading the discussion in the text, explain exactly how the total of* 26 *or* 28 ATP *from oxidative phosphorylation was calculated (see the yellow bar).*

Animation

ATP Yield from Cellular Respiration

image

image

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Why are the numbers in Figure 9.15 inexact? There are three reasons we cannot state an exact number of ATP molecules generated by the breakdown of one molecule of glucose. First, phosphorylation and the redox reactions are not directly coupled to each other, so the ratio of the number of NADH molecules to the number of ATP molecules is not a whole number. We know that 1 NADH results in 10 H+ being transported out across the inner mitochondrial membrane, but the exact number of H+ that must reenter the mitochondrial matrix via ATP synthase to generate 1 ATP has long been debated. Based on experimental data, however, most biochemists now agree that the most accurate number is 4 H+. Therefore, a single molecule of NADH generates enough proton-motive force for the synthesis of 2.5 ATP. The citric acid cycle also supplies electrons to the electron transport chain via FADH2, but since its electrons enter later in the chain, each molecule of this electron carrier is responsible for transport of only enough H+ for the synthesis of 1.5 ATP. These numbers also take into account the slight energetic cost of moving the ATP formed in the mitochondrion out into the cytosol, where it will be used.

Second, the ATP yield varies slightly depending on the type of shuttle used to transport electrons from the cytosol into the mitochondrion. The mitochondrial inner membrane is not permeable to NADH, so NADH in the cytosol is segregated from the machinery of oxidative phosphorylation. The 2 electrons of NADH captured in glycolysis must be conveyed into the mitochondrion by one of several electron shuttle systems. Depending on the kind of shuttle in a particular cell type, the electrons are passed either to NAD+ or to FAD

in the mitochondrial matrix (see Figures 9.14 and 9.15). If the electrons are passed to FAD, as in brain cells, only about 1.5 ATP can result from each NADH that was originally generated in the cytosol. If the electrons are passed to mitochondrial NAD+, as in liver cells and heart cells, the yield is about 2.5 ATP per NADH.

A third variable that reduces the yield of ATP is the use of the proton-motive force generated by the redox reactions of respiration to drive other kinds of work. For example, the proton-motive force powers the mitochondrion's uptake of pyruvate from the cytosol (see Figure 9.9). However, if *all* the proton-motive force generated by the electron transport chain were used to drive ATP synthesis, one glucose molecule could generate a maximum of 28 ATP produced by oxidative phosphorylation plus 4 ATP (net) from substrate-level phosphorylation to give a total yield of about 32 ATP (or only about 30 ATP if the less efficient shuttle were functioning).

Animation

Electron Transport Chain: Factors Affecting ATP Yield

image

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We can now roughly estimate the efficiency of respiration—that is, the percentage of chemical energy in glucose that has been transferred to ATP. Recall that the complete oxidation of a mole of glucose releases 686 kcal of energy under standard conditions ($\Delta G = -686$ kcal/mol). Phosphorylation of ADP to form ATP stores at least 7.3 kcal per mole of ATP. Therefore, the efficiency of respiration is 7.3 kcal per mole of ATP times 32 moles of ATP per mole of glucose divided by 68 kcal per mole of glucose, which equals 0.34. Thus, about 34% of the potential chemical energy in glucose has been transferred to ATP; the actual percentage is bound to vary as ΔG varies under different cellular conditions. Cellular respiration is remarkably efficient in its energy conversion. By comparison, even the most efficient automobile converts only about 25% of the energy stored in gasoline to energy that moves the car.

The rest of the energy stored in glucose is lost as heat. We humans use some of this heat to maintain our relatively high body temperature (37°C), and we dissipate the rest through sweating and other cooling mechanisms.

Surprisingly, perhaps, it may be beneficial under certain conditions to reduce the efficiency of cellular respiration. A remarkable adaptation is shown by hibernating mammals, which overwinter in a state of inactivity and lowered metabolism. Although their internal body temperature is lower than normal, it still must be kept significantly higher than the external air temperature. One type of tissue, called brown fat, is made up of cells packed full of mitochondria. The inner mitochondrial membrane contains a channel protein called the uncoupling protein that allows protons to flow back down their concentration gradient without

generating ATP. Activation of these proteins in hibernating mammals results in ongoing oxidation of stored fuel (fats), generating heat without any ATP production. In the absence of such an adaptation, the buildup of ATP would eventually cause cellular respiration to be shut down by regulatory mechanisms that will be discussed later. Brown fat is also used for heat generation in humans. In the **Scientific Skills Exercise**, you can work with data in a related but different case where a decrease in metabolic efficiency in cells is used to generate heat.

Scientific Skills Exercise

Making a Bar Graph and Evaluating a Hypothesis

AP Science Practices 1, 2, 3, 4

Does Thyroid Hormone Level Affect O_2 **Consumption in Cells?** Some animals, such as mammals and birds, maintain a relatively constant body temperature, above that of their environment, by using heat produced as a by-product of metabolism. When the core temperature of these animals drops below an internal set point, their cells are triggered to reduce the efficiency of ATP production by the electron transport chains in mitochondria. At lower efficiency, extra fuel must be consumed to produce the same number of ATPs, generating additional heat. This response is moderated by the endocrine system, and researchers hypothesized that it might be triggered by thyroid hormone. In this exercise, you will use a bar graph to visualize data from an experiment that compared the metabolic rates (by measuring O_2 consumption) in mitochondria of cells from animals with different levels of thyroid hormone.

A mouse.

How the Experiment Was Done Liver cells were isolated from sibling rats that had low, normal, or elevated thyroid hormone levels. The oxygen consumption rate due to activity of the mitochondrial electron transport chains of each type of cell was measured under controlled conditions.

Data from the Experiment

A table lists thyroid hormone levels and oxygen consumption rate.

Data from M. E. Harper and M. D. Brand, The quantitative contributions of mitochondrial proton leak and ATP turnover reactions to the changed respiration rates of hepatocytes from rats of different thyroid status, *Journal of Biological Chemistry* 268:14850–14860 (1993).

Thyroid hormone level	Oxygen consumption rate
Low	4.3
Normal	4.8
Elevated	8.7

INTERPRET THE DATA

- To visualize any differences in O2 consumption between cell types, it will be useful to graph the data in a bar graph. First, set up the axes.
- What is the independent variable (intentionally varied by the researchers), which goes on the x-axis? List the categories along the x-axis; because they are discrete rather than continuous, you can list them in any order.
- What is the dependent variable (measured by the researchers), which goes on the y-axis?
- What units (abbreviated) should go on the y-axis? Label the y-axis, including the units specified in the data table. Determine the range of values of the data that will need to go on the y-axis. What is the largest value? Draw evenly spaced tick marks and label them, starting with 0 at the bottom.
- Graph the data for each sample. Match each x-value with its y-value and place a mark on the graph at that coordinate, then draw a bar from the x-axis up to the correct height for each sample. Why is a bar graph more appropriate than a scatter plot or line graph? (For additional information about graphs, see the Scientific Skills Review in Appendix D.)
- Examine your graph and look for a pattern in the data.
- Which cell type had the highest rate of O2 consumption, and which had the lowest?
- Does this support the researchers' hypothesis? Explain.
- Based on what you know about mitochondrial electron transport and heat production, predict which rats had the highest, and which had the lowest, body temperature.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

BioFlix® Animation

Cellular Respiration

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Concept 9.5: Fermentation and anaerobic respiration enable cells to produce ATP without the use of oxygen

Because most of the ATP generated by cellular respiration is due to the work of oxidative phosphorylation, our estimate of ATP yield from aerobic respiration depends on an adequate supply of O_2 to the cell. Without the electronegative oxygen atoms in O_2 to pull electrons down the transport chain, oxidative phosphorylation eventually ceases. However, there are two general mechanisms by which certain cells can oxidize organic fuel and generate ATP *without* the use of O_2: anaerobic respiration and fermentation. The distinction between these two is that an electron transport chain is used in anaerobic respiration but not in fermentation. (The electron transport chain is also called the respiratory chain because of its role in both types of cellular respiration.)

We have already mentioned anaerobic respiration, which takes place in certain prokaryotic organisms that live in environments without O_2. These organisms have an electron transport chain but do not use O_2 as a final electron acceptor at the end of the chain. O_2 performs this function very well because it consists of two extremely electronegative atoms, but other substances can also serve as final electron acceptors. Some "sulfate-reducing" marine bacteria, for instance, use the sulfate ion (SO_4^{2-}) at the end of their respiratory chain. Operation of the chain builds up a proton-motive force used to produce ATP, but H_2S (hydrogen sulfide) is made as a by-product rather than water. The rotten-egg odor you may have smelled while walking through a salt marsh or a mudflat signals the presence of sulfate-reducing bacteria.

Fermentation is a way of harvesting chemical energy without using either O_2 or any electron transport chain—in other words, without cellular respiration. How can food be oxidized without cellular respiration? Remember, oxidation simply refers to the loss of electrons to an electron acceptor, so it does not need to involve O_2. Glycolysis oxidizes glucose to two molecules of pyruvate. The oxidizing agent of glycolysis is NAD^+, and neither O_2 nor any electron transfer chain is involved. Overall, glycolysis is exergonic, and some of the energy made available is used to produce 2 ATP (net) by substrate-level phosphorylation. If O_2 *is* present, then additional ATP is made by oxidative phosphorylation when NADH passes electrons removed from glucose to the electron transport chain. But glycolysis generates 2 ATP whether oxygen is present or not—that is, whether conditions are aerobic or anaerobic.

As an alternative to respiratory oxidation of organic nutrients, fermentation is an extension of glycolysis that allows continuous generation of ATP by the substrate-level phosphorylation of glycolysis. For this to occur, there must be a sufficient supply of NAD+ to accept electrons during the oxidation step of glycolysis. Without some mechanism to recycle NAD+ from NADH, glycolysis would soon deplete the cell's pool of NAD+ by reducing it all to NADH and would shut itself down for lack of an oxidizing agent. Under aerobic conditions, NAD+ is recycled from NADH by the transfer of electrons to the electron transport chain. An anaerobic alternative is to transfer electrons from NADH to pyruvate, the end product of glycolysis.

Types of Fermentation

Fermentation consists of glycolysis plus reactions that regenerate NAD+ by transferring electrons from NADH to pyruvate or derivatives of pyruvate. The NAD+ can then be reused to oxidize sugar by glycolysis, which nets two molecules of ATP by substrate-level phosphorylation. There are many types of fermentation, differing in the end products formed from pyruvate. Two types are alcohol fermentation and lactic acid fermentation, and both are harnessed by humans for food and industrial production.

In **alcohol fermentation(Figure 9.16a)**, pyruvate is converted to ethanol (ethyl alcohol) in two steps. The first step releases CO_2 from the pyruvate, which is converted to the two-carbon compound acetaldehyde. In the second step, acetaldehyde is reduced by NADH to ethanol. This regenerates the supply of NAD+ needed for the continuation of glycolysis. Many bacteria carry out alcohol fermentation under anaerobic conditions. Yeast (a fungus), in addition to aerobic respiration, also carries out alcohol fermentation. For thousands of years, humans have used yeast in brewing, winemaking, and baking. The CO_2 bubbles generated by baker's yeast during alcohol fermentation allow bread to rise.

Figure 9.16

Fermentation.

A flowchart illustrating the process of Fermentation in two parts a and b.

Part a. Alcohol fermentation. Glucose goes to pyruvate during glycolysis, forming 2 A T P. Pyruvate goes to acetaldehyde, which goes to 2 ethanol, forming 2 N A D plus from 2 N A D H plus 2 H plus. During glycolysis, glucose goes to 2 pyruvates (the structure is a 3-carbon chain with the first carbon in a carboxyl group, the second carbon in a carbonyl group, and the third attached to H 3). In the process, 2 A T P is made from 2 A D P plus 2 inorganic phosphates. Next, 2 pyruvates go to 2 acetaldehyde, giving off 2 C O 2 in the process (the carbons came from the carboxyl groups of pyruvate). The structure of acetaldehyde is a 2-carbon chain with the first carbon in a carbonyl group attached to H and the second bonded to H 3. Then, 2 acetaldehyde goes to 2 ethanol, and 2 N A D plus are formed from 2 N A D H plus 2 H plus, which can cycle

back to 2 N A D H plus 2 H plus. The structure of ethanol is a central carbon bonded to 2 H's, an O H, and a C H 3.

Part b. Lactic acid fermentation. Glucose goes to 2 pyruvates during glycolysis, forming 2 A T P. 2 Pyruvate goes to 2 lactate, forming 2 N A D plus from 2 N A D H plus 2 H plus. Glucose goes to 2 pyruvate through glycolysis, and 2 A T P are formed from 2 A D P plus 2 inorganic phosphates. 2 pyruvate goes to 2 lactate, and 2 N A D plus are formed from 2 N A D H plus 2 H plus, which can cycle back to 2 N A D H plus 2 H plus. The structure of lactate is a 3-carbon chain, with the first carbon in a carboxyl group, the second attached to an O H and an H, and the third attached to H 3.

In the absence of oxygen, many cells use fermentation to produce ATP by substrate-level phosphorylation. NAD+ is regenerated for use in glycolysis when pyruvate, the end product of glycolysis, serves as an electron acceptor for oxidizing NADH. Two of the common end products formed from fermentation are **(a)** ethanol and **(b)** lactate, the ionized form of lactic acid.

Animation

Fermentation

image

Play

image

Play

**

During **lactic acid fermentation(Figure 9.16b)**, pyruvate is reduced directly by NADH to form lactate as an end product, regenerating NAD+ with no release of CO_2. (Lactate is the ionized form of lactic acid.) Lactic acid fermentation by certain fungi and bacteria is used in the dairy industry to make cheese and yogurt. A complex series of fermentation and aerobic respiration pathways carried out by yeasts and bacteria on cacao beans is responsible for the production of chocolate.

Cacao beans and fruits.

Cacao beans and fruits

What about lactate production in humans? Previously, we thought that human muscle cells only produced lactate when O_2 was in short supply, such as during intense exercise. Research done over the last few decades, though, indicates that the lactate story, in mammals at least, is more complicated. There are two types of skeletal muscle fibers. One (red muscle) preferentially oxidizes glucose completely to CO_2; the other

(white muscle) produces significant amounts of lactate from the pyruvate made during glycolysis, even under aerobic conditions, offering fast but energetically inefficient ATP production. The lactate product is then mostly oxidized by red muscle cells in the vicinity, with the remainder exported to liver or kidney cells for glucose formation. Because this lactate production is not anaerobic, but the result of glycolysis in these cells, exercise physiologists prefer not to use the term “fermentation.”

During strenuous exercise, when carbohydrate catabolism outpaces the supply of O_2 from the blood to the muscle, lactate can’t be oxidized to pyruvate. The lactate that accumulates was once thought to cause muscle fatigue during intense exercise and pain a day or so later. However, research suggests that, contrary to popular opinion, lactate production actually improves performance during exercise! Furthermore, within an hour, excess lactate is shuttled to other tissues for oxidation or to the liver and kidneys for production of glucose or its storage molecule, glycogen. (Nextday muscle soreness is more likely caused by trauma to cells in small muscle fibers, which leads to inflammation and pain.)

Comparing Fermentation with Anaerobic and Aerobic Respiration

Fermentation, anaerobic respiration, and aerobic respiration are three alternative cellular pathways for producing ATP by harvesting the chemical energy of food. All three use glycolysis to oxidize glucose and other organic fuels to pyruvate, with a net production of 2 ATP by substrate-level phosphorylation. And in all three pathways, NAD^+ is the oxidizing agent that accepts electrons from food during glycolysis.

A key difference is the contrasting mechanisms for oxidizing NADH back to NAD^+, which is required to sustain glycolysis. In fermentation, the final electron acceptor is an organic molecule such as pyruvate (lactic acid fermentation) or acetaldehyde (alcohol fermentation). In cellular respiration, by contrast, electrons carried by NADH are transferred to an electron transport chain, which regenerates the NAD^+ required for glycolysis.

Another major difference is the amount of ATP produced. Fermentation yields two molecules of ATP, produced by substrate-level phosphorylation. In the absence of an electron transport chain, the energy stored in pyruvate is unavailable. In cellular respiration, however, pyruvate is completely oxidized in the mitochondrion. Most of the chemical energy from this process is shuttled by NADH and $FADH_2$ in the form of electrons to the electron transport chain. There, the electrons move stepwise down a series of redox reactions to a final electron acceptor. (In aerobic respiration, the final electron acceptor is O_2; in anaerobic respiration, the final acceptor is another molecule with a high affinity for electrons (and thus a good electron acceptor), although less so than O_2.) Stepwise electron transport drives oxidative phosphorylation, yielding ATP. Thus, cellular respiration harvests much more energy from each sugar molecule than fermentation can.

In fact, aerobic respiration yields up to 32 molecules of ATP per glucose molecule—up to 16 times as much as does fermentation.

Some organisms, called **obligate anaerobes**, carry out only fermentation or anaerobic respiration. In fact, these organisms cannot survive in the presence of oxygen, some forms of which can actually be toxic if protective systems are not present in the cell. A few cell types, such as cells of the vertebrate brain, can carry out only aerobic oxidation of pyruvate, and need O2 to survive. Other organisms, including yeasts and many bacteria, can make enough ATP to survive using either fermentation or respiration. Such species are called **facultative anaerobes**. In yeast cells, for example, pyruvate is a fork in the metabolic road that leads to two alternative catabolic routes **(Figure 9.17)**. Under aerobic conditions, pyruvate can be converted to acetyl CoA, and oxidation continues in the citric acid cycle via aerobic respiration. Under anaerobic conditions, lactic acid fermentation occurs. Pyruvate is diverted from the citric acid cycle, serving instead as an electron acceptor to recycle NAD+. To make the same amount of ATP, a facultative anaerobe has to consume sugar at a much faster rate when fermenting than when respiring.

Figure 9.17

Pyruvate as a key juncture in catabolism.

Pyruvate as a key juncture in catabolism.

Glycolysis is common to fermentation and cellular respiration. The end product of glycolysis, pyruvate, represents a fork in the catabolic pathways of glucose oxidation. In facultative anaerobes, such as yeasts, which are capable of both aerobic cellular respiration and fermentation, pyruvate is committed to one of those two pathways, usually depending on whether or not oxygen is present.

Glucose is converted to pyruvate during glycolysis in the cytosol, after which either fermentation (no O 2 present) or aerobic cellular respiration (O 2 present) can occur. The products of fermentation are ethanol, lactate, or other products. In aerobic cellular respiration, pyruvate is converted to Acetyl Co A in the mitochondrion, which enters the Citric Acid Cycle.

The Evolutionary Significance of Glycolysis

EVOLUTION The role of glycolysis in both fermentation and respiration has an evolutionary basis. Ancient prokaryotes are thought to have used glycolysis to make ATP long before oxygen was present in Earth's atmosphere. The oldest known fossils of bacteria date back 3.5 billion years, but appreciable quantities of oxygen probably did not begin to accumulate in the atmosphere until about 2.7 billion years ago. Cyanobacteria produced this O2 as a by-product of photosynthesis. Therefore, early prokaryotes may have generated ATP exclusively from glycolysis. The fact that glycolysis is today the most widespread metabolic

pathway among Earth's organisms suggests that it evolved very early in the history of life. The cytosolic location of glycolysis also implies great antiquity; the pathway does not require any of the membrane-enclosed organelles of the eukaryotic cell, which evolved approximately 1 billion years after the first prokaryotic cell. Glycolysis is a metabolic heirloom from early cells that continues to function in fermentation and as the first stage in the breakdown of organic molecules by respiration.

Concept 9.6: Glycolysis and the citric acid cycle connect to many other metabolic pathways

So far, we have looked at the oxidative breakdown of glucose in isolation from the cell's overall metabolic economy. In this section, you will learn that glycolysis and the citric acid cycle are major intersections of the cell's catabolic (breakdown) and anabolic (biosynthetic) pathways.

The Versatility of Catabolism

Throughout this chapter, glucose has been used as an example of a fuel for cellular respiration. But free glucose molecules are not common in the diets of humans and other animals. We obtain most of our calories in the form of fats, proteins, and carbohydrates such as sucrose and other disaccharides, and starch, a polysaccharide. All these organic molecules in food can be used by cellular respiration to make ATP **(Figure 9.18)**.

Figure 9.18

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The catabolism of various molecules from food.
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The catabolism of various molecules from food.

Carbohydrates, fats, and proteins can all be used as fuel for cellular respiration. Monomers of these molecules enter glycolysis or the citric acid cycle at various points. Glycolysis and the citric acid cycle are catabolic funnels through which electrons from all kinds of organic molecules flow on their exergonic fall to oxygen.

Proteins are broken down into amino acids, which must then have an N H 3 removed before entry into the cellular respiration pathway as pyruvate during glycolysis, as Acetyl Co A, or directly enter the Citric Acid Cycle. Carbohydrates are broken down into sugars which can then enter glycolysis. Fats are broken down into glycerol and fatty acids. Glycerol can enter glycolysis as glyceraldehyde 3-Phosphate; fatty acids enter the Citric Acid Cycle as Acetyl Co A.

Glycolysis can accept a wide range of carbohydrates for catabolism. In the digestive tract, starch is hydrolyzed to glucose, which is broken down in cells by glycolysis and the citric acid cycle. Glycogen, the polysaccharide that humans and many other animals store in their liver and muscle cells, can be hydrolyzed to glucose between meals as fuel for respiration. Digestion of disaccharides, including sucrose, provides glucose and other monosaccharides as fuel for respiration.

Proteins can also be used for fuel, but first they must be digested to their constituent amino acids. Many of the amino acids are used by the organism to build new proteins. Amino acids present in excess are converted by enzymes to intermediates of glycolysis and the citric acid cycle. Before amino acids can feed into glycolysis or the citric acid cycle, their amino groups must be removed, a process called *deamination*. The nitrogenous waste is excreted from the animal in the form of ammonia NH_3, urea, or other waste products.

Catabolism can also harvest energy stored in fats obtained either from food or from fat cells in the body. After fats are digested to glycerol and fatty acids, the glycerol is converted to glyceraldehyde 3-phosphate, an intermediate of glycolysis. Most of the energy of a fat is stored in the fatty acids. A metabolic sequence called **beta oxidation** breaks the fatty acids down to two-carbon fragments, which enter the citric acid cycle as acetyl CoA. NADH and $FADH_2$ are also generated during beta oxidation; they can enter the electron transport chain, leading to further ATP production. Fats make excellent fuels, in large part due to their chemical structure and the high energy level of their electrons (present in many C—H bonds, equally shared between C and H) compared to those of carbohydrates. A gram of fat oxidized by respiration produces more than twice as much ATP as a gram of carbohydrate. Unfortunately, this also means that a person trying to lose weight must work hard to use up fat stored in the body because so many calories are stockpiled in each gram of fat.

Biosynthesis (Anabolic Pathways)

Cells need substance as well as energy. Not all the organic molecules of food are destined to be oxidized as fuel to make ATP. In addition to calories, food must also provide the carbon skeletons that cells require to make their own molecules. Some organic monomers obtained from digestion can be used directly. For example, as previously mentioned, amino acids from the hydrolysis of proteins in food can be incorporated into the organism’s own proteins. Often, however, the body needs specific molecules that are not present as

such in food. Compounds formed as intermediates of glycolysis and the citric acid cycle can be diverted into anabolic pathways as precursors from which the cell can synthesize the molecules it requires. For example, humans can make about half of the 20 amino acids in proteins by modifying compounds siphoned away from the citric acid cycle; the rest are “essential amino acids” that must be obtained in the diet. Also, glucose can be made from pyruvate, and fatty acids can be synthesized from acetyl CoA. Of course, these anabolic, or biosynthetic, pathways do not generate ATP, but instead consume it.

In addition, glycolysis and the citric acid cycle function as metabolic interchanges that enable our cells to convert some kinds of molecules to others as we need them. For example, an intermediate compound generated during glycolysis, dihydroxyacetone phosphate (see Figure 9.8, step 5), can be converted to one of the major precursors of fats. If we eat more food than we need, we store fat even if our diet is fat-free. Metabolism is remarkably versatile and adaptable.

Regulation of Cellular Respiration via Feedback Mechanisms

Basic principles of supply and demand regulate the metabolic economy. The cell does not waste energy making more of a particular substance than it needs. If there is a surplus of a certain amino acid, for example, the anabolic pathway that synthesizes that amino acid from an intermediate of the citric acid cycle is switched off. The most common mechanism for this control is feedback inhibition: The end product of the anabolic pathway inhibits the enzyme that catalyzes an early step of the pathway (see Figure 8.21). This prevents the needless diversion of key metabolic intermediates from uses that are more urgent.

The cell also controls its catabolism. If the cell is working hard and its ATP concentration begins to drop, cellular respiration speeds up. When there is plenty of ATP to meet demand, respiration slows down, sparing valuable organic molecules for other functions. Again, control is based mainly on regulating the activity of enzymes at strategic points in the catabolic pathway. As shown in **Figure 9.19**, one important switch is phosphofructokinase, the enzyme that catalyzes step 3 of glycolysis (see Figure 9.8). That is the first step that commits the substrate irreversibly to the glycolytic pathway. By controlling the rate of this step, the cell can speed up or slow down the entire catabolic process. Phosphofructokinase can thus be considered the pacemaker of cellular respiration.

Figure 9.19

The control of cellular respiration.

A T P made during glycolysis, the citric acid cycle, and oxidative phosphorylation inhibits phosphofructokinase, as does citrate produced during the citric acid cycle. AMP stimulates phosphofructokinase.

During glycolysis, glucose is converted to Fructose 6-phosphate, and then to Fructose 1,6-biphosphate with the allosteric enzyme phosphofructokinase. After a series of reactions, Fructose 1,6-biphosphate is converted to pyruvate, forming A T P in the process. Pyruvate is converted to Acetyl Co A, which enters the Citric Acid Cycle, the products of which go on to oxidative phosphorylation. Both of these steps also result in the formation of A T P. Both A T P and citrate produced during the Citric Acid Cycle inhibit phosphofructokinase, preventing the progression of the pathway. A M P (derived from A D P) stimulates phosphofructokinase.

Allosteric enzymes at certain points in the respiratory pathway respond to inhibitors and activators that help set the pace of glycolysis and the citric acid cycle. Phosphofructokinase, which catalyzes an early step in glycolysis (see Figure 9.8, **STEP 3** and Figure 6.32b), is one such enzyme. It is stimulated by AMP (derived from ADP) but is inhibited by ATP and by citrate. This feedback regulation adjusts the rate of respiration as the cell's catabolic and anabolic demands change.

Phosphofructokinase is an allosteric enzyme with receptor sites for specific inhibitors and activators. It is inhibited by ATP and stimulated by AMP (adenosine monophosphate), which the cell derives from ADP. As ATP accumulates, inhibition of the enzyme slows down glycolysis. The enzyme becomes active again as cellular work converts ATP to ADP (and AMP) faster than ATP is being regenerated. Phosphofructokinase is also sensitive to citrate, the first product of the citric acid cycle. If citrate accumulates in mitochondria, some of it passes into the cytosol and inhibits phosphofructokinase. This mechanism helps synchronize the rates of glycolysis and the citric acid cycle. As citrate accumulates, glycolysis slows down, and the supply of pyruvate and thus acetyl groups to the citric acid cycle decreases. If citrate consumption increases, either because of a demand for more ATP or because anabolic pathways are draining off intermediates of the citric acid cycle, glycolysis accelerates and meets the demand. Metabolic balance is augmented by the control of enzymes that catalyze other key steps of glycolysis and the citric acid cycle. Cells are thrifty, expedient, and responsive in their metabolism.

Review the first page of this chapter to put cellular respiration into the broader context of energy flow and chemical cycling in ecosystems. The energy that keeps us alive is *released*, not *produced*, by cellular respiration. We are tapping energy that was stored in food by photosynthesis, which captures light and converts it to chemical energy, a process you will learn about next, in Chapter 10.